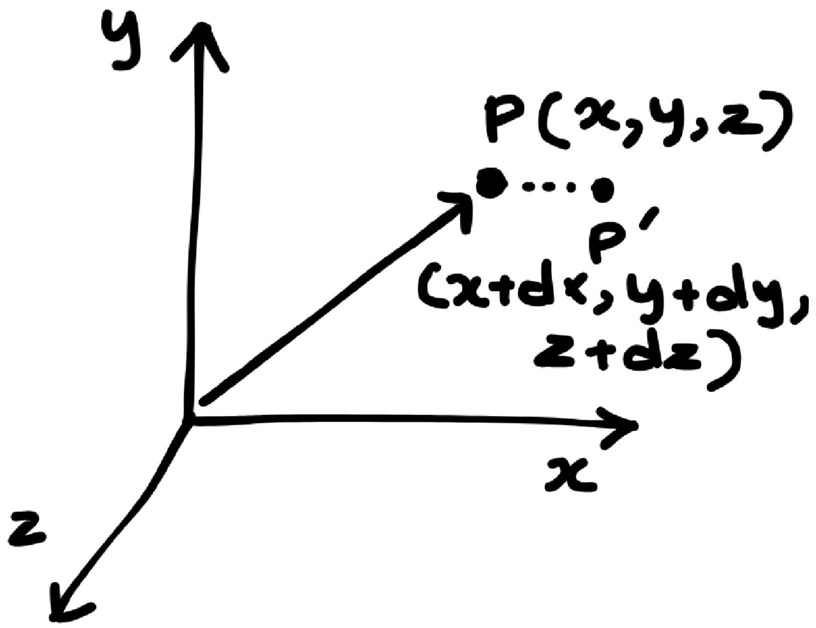
u2-electromagnetic theory

n01-coordinate_system_and_position_vectors



$$P(x, y, z)$$

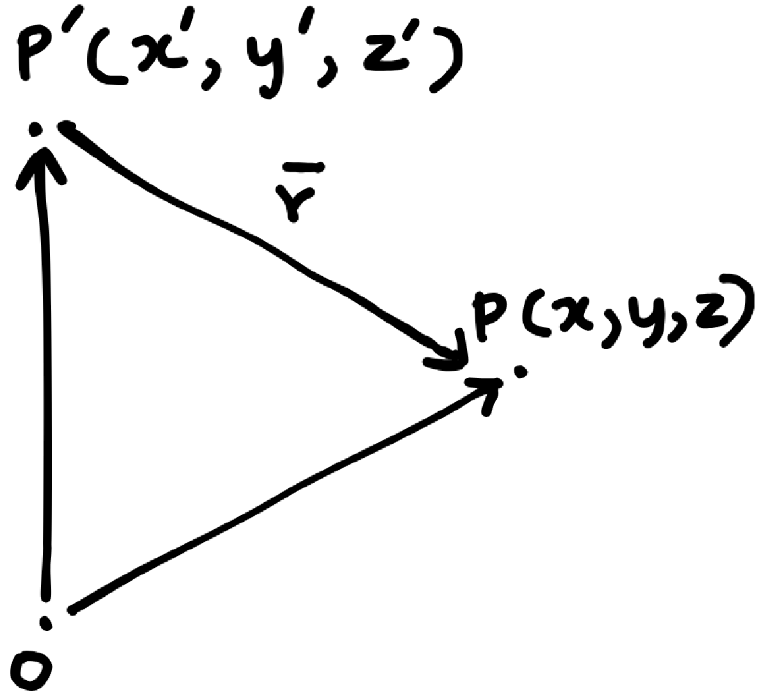
$$\vec{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$d\vec{r} = d\vec{l} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$$

- **Field point**: the point where the electric field has to be determined.
- **Source point**: the point where the electric charge is located.
- The **position vector** $\vec{r}$ is given by the **difference of source point and field point**.



$$\vec{r} = (x - x')\hat{i} + (y - y')\hat{j} + (z - z')\hat{k}$$

The differential of a scalar function T:

$$dT = \left(\frac{\partial T}{\partial x}\right)dx + \left(\frac{\partial T}{\partial y}\right)dy + \left(\frac{\partial T}{\partial z}\right)dz$$

$$dT = \left(\frac{\partial T}{\partial x}\hat{i} + \frac{\partial T}{\partial y}\hat{j} + \frac{\partial T}{\partial z}\hat{k}\right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$dT = (\nabla T) \cdot (d\vec{l})$$

In the above equation,

$$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

is called the del operator which can act on both scalar or vector form.

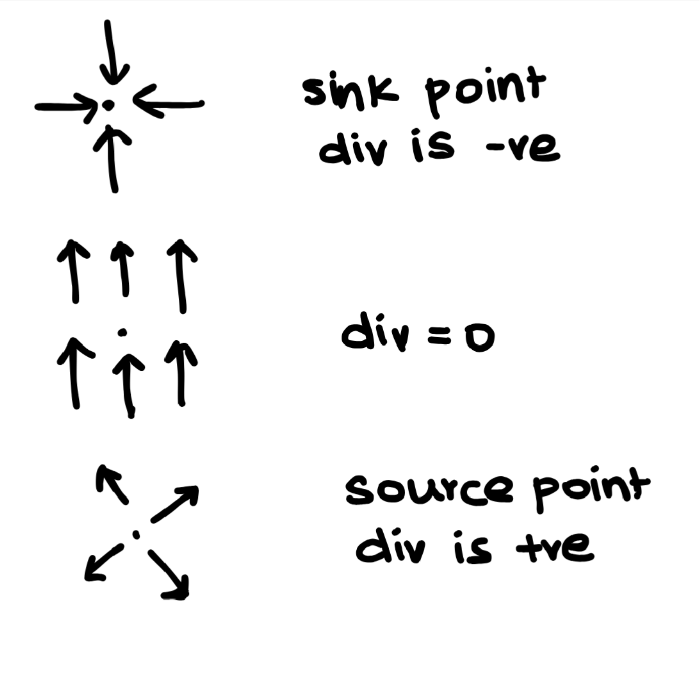
- Over a scalar: $\nabla T \rightarrow$ **gradient** of T
- Over a vector as dot product: $\nabla \cdot \vec{V} \rightarrow$ **divergence** of $\vec{V}$
- Over a vector via cross product: $\nabla \times \vec{V} \rightarrow$ **curl** of $\vec{V}$

n02-divergence

Divergence ($\nabla \cdot \vec{V}$):

For any vector $\vec{V}$, $\nabla \cdot \vec{V}$ is called the **divergence of** $\vec{V}$ and it is denoted by $\operatorname{div} \vec{V}$.

- $\operatorname{div} \vec{V}$ is an **scalar quantity**.
- Geometrically, $\nabla \cdot \vec{V}$ at any given point is a measure of how much the vector $\vec{V}$ spreads out from that point.



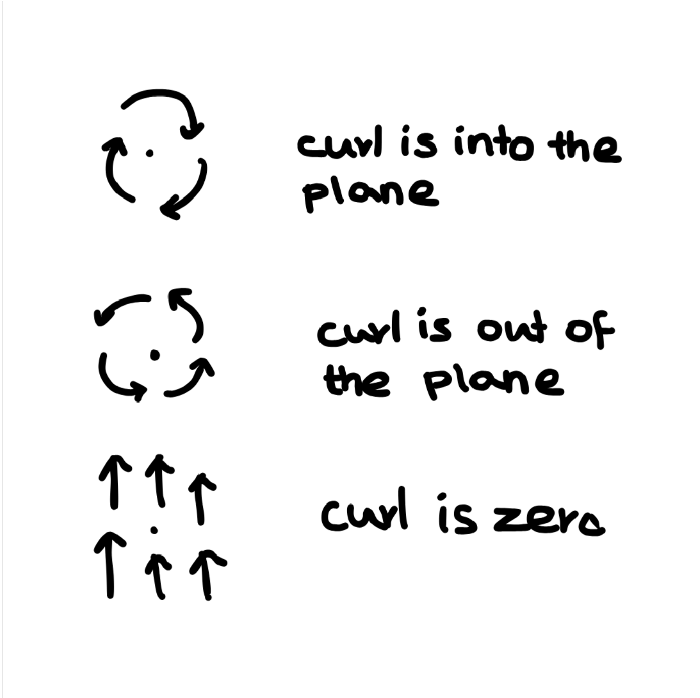
- **Properties**:
  - produces a scalar quantity
  - divergence of a scalar makes no sense.
  - $\vec{\nabla} \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$
  - $\nabla \cdot (V\vec{A}) = V(\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla} V)$

n03-curl

Curl($\nabla \times \vec{V}$):

For any given vector $\vec{V}$, $\nabla \times \vec{V}$ is known as the **curl of a vector**.

- Curl $\vec{V}$ is always a **vector**.
- Geometrically, $\operatorname{curl} \vec{V}$ is the measure of how much the vector $\vec{V}$ spins around the given point.
- The curl provides the **maximum value of circulation** of $\vec{V}$ per unit area and indicates the direction along which this maximum value occurs.



Properties:

- $\nabla \times (\vec{A} + \vec{B}) = (\nabla \times \vec{A}) + (\nabla \times \vec{B})$
- $\nabla \times (\nabla V) = 0$ **(curl of gradient)**
- $\nabla \cdot (\nabla \times V) = 0$ (div of curl)
- $\nabla \times (V\vec{A}) = V(\nabla \times \vec{A}) + (\nabla V) \times \vec{A}$

n04-integral_theorems_and_integrals

Stokes' Theorem (Fundamental theorem for Curl):

It is mathematically given as:

$${}_{P}\vec{V} \cdot d\vec{l} = \int (\nabla \times \vec{V}) \cdot d\vec{a}$$

The **circulation** of a vector function $\vec{V}$ around a **closed path** P is equal to the **surface integral of curl of** $\vec{V}$ over the open surface bounded by P, provided $\vec{V}$ and $\nabla \times \vec{V}$ are continuous on .
where the directions of $d\vec{l}$ and $d\vec{a}$ by using right hand thumb rule.

Gauss Divergence Theorem:

It states that the **net outward flux** of a vector function $\vec{V}$ through the **closed surface** is same as the **volume integral of the divergence of** $\vec{V}$.
The mathematical notation is:

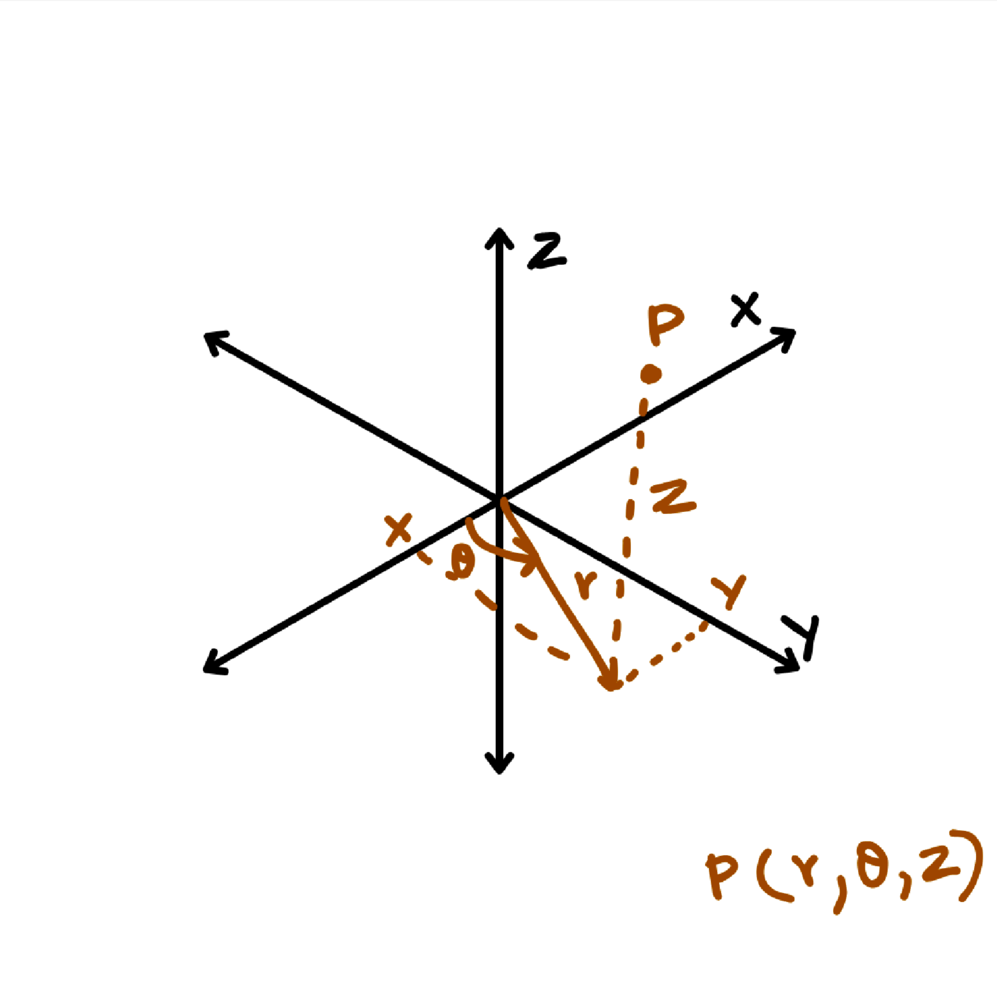
$$\vec{V} \cdot d\vec{a} = \int_V (\nabla \cdot \vec{V}) d\tau$$

Integrals:

- Line integral: $\int_a f(x)dl$
- Areal integral: $f \cdot d\vec{a} = fdxdy + fdydz + fdzdx$
- Volume integral: $fdV = fdxdydz$

n05-cylindrical_coordinates

visualisation



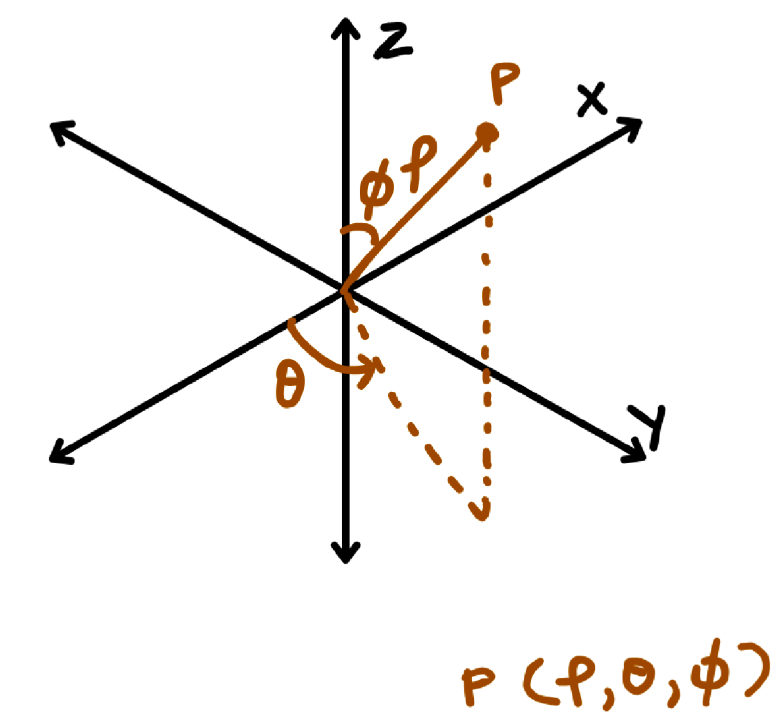
cylindrical to rectangular coordinates

- $x = r\cos\theta$
- $y = r\sin\theta$
- $z = z$

rectangular to cylindrical coordinates

- $r^2 = x^2 + y^2$
- $\tan\theta = \frac{y}{x}$
- $z = z$

n06-spherical_coordinates



spherical to rectangular

- $x = \rho \sin\phi \cos\theta$
- $y = \rho \sin\phi \sin\theta$
- $z = \rho \cos\phi$

rectangular to spherical

- $\rho = \sqrt{x^2 + y^2 + z^2}$
- $\tan\theta = \frac{y}{x}$
- $\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}}$

n07-coulomb's_law

$$\vec{F} = \frac{1}{4\pi_0} \frac{q_1 q_2}{r^2} \hat{r}$$

- Now, suppose if there are many charges $q_1, q_2, \ldots, q_n$. Then the force exerted due to all these charges at is given by **Principle of Superposition**.

  $$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n$$

  where $\vec{F}_i$ is the force at due exerted by q_i along by ignoring all other charges.

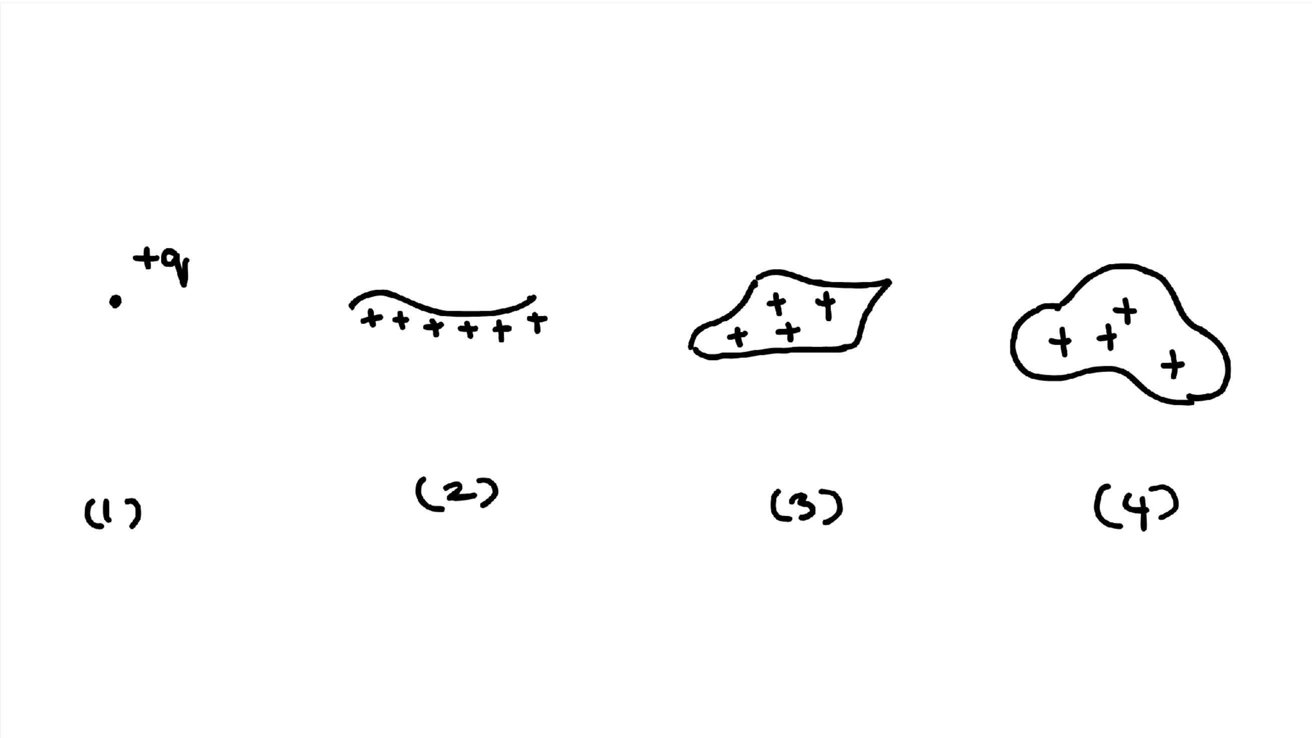
$$\vec{F}_{\text{net}} = \frac{1}{4\pi_0}\frac{q_1}{r_1^2}\hat{r}_1 + \frac{1}{4\pi_0}\frac{q_2}{r_2^2}\hat{r}_2 + \cdots + \frac{1}{4\pi_0}\frac{q_n}{r_n^2}\hat{r}_n$$

$$= \frac{}{4\pi_0} \; {}^{n}{}_{i=1} \frac{q_i}{r_i^2}\hat{r}_i$$

$$\vec{F}_{\text{net}} = \vec{E}_{\text{net}}$$

$$\vec{F}_{\text{net}} = \frac{1}{4\pi_0} \; {}^{n}{}_{i=1} \frac{q_i}{r_i^2}\hat{r}_i$$

n08-electric_field_due_to_continuous_charge_distribution



For **continuous charge distribution**, q is replaced by integral.

1. If the charge is distributed along a line, as shown in figure (2), with charge per unit length = λ (Linear charge density). Then

$$dq = \lambda dl$$

2. If the charge is distributed across a surface as shown in figure (3) σ = charge per unit area (Surface charge density). Then

$$dq = \sigma dA$$

3. If the charge fills a volume, with ρ as charge per unit volume (Volume charge density) (figure 4). ρ = Volume charge density. Then

$$dq = \rho dV$$

The electric field due to different charge distributions:

1. **Point Charge**:

$$d\vec{E} = \frac{1}{4\pi_0} \frac{q}{r^2} \hat{r}$$

2. **Line Charge**:

$$\int d\vec{E} = \int \frac{1}{4\pi_0} \frac{\lambda dl}{r^2} \hat{r}$$

3. **Areal Charge**:

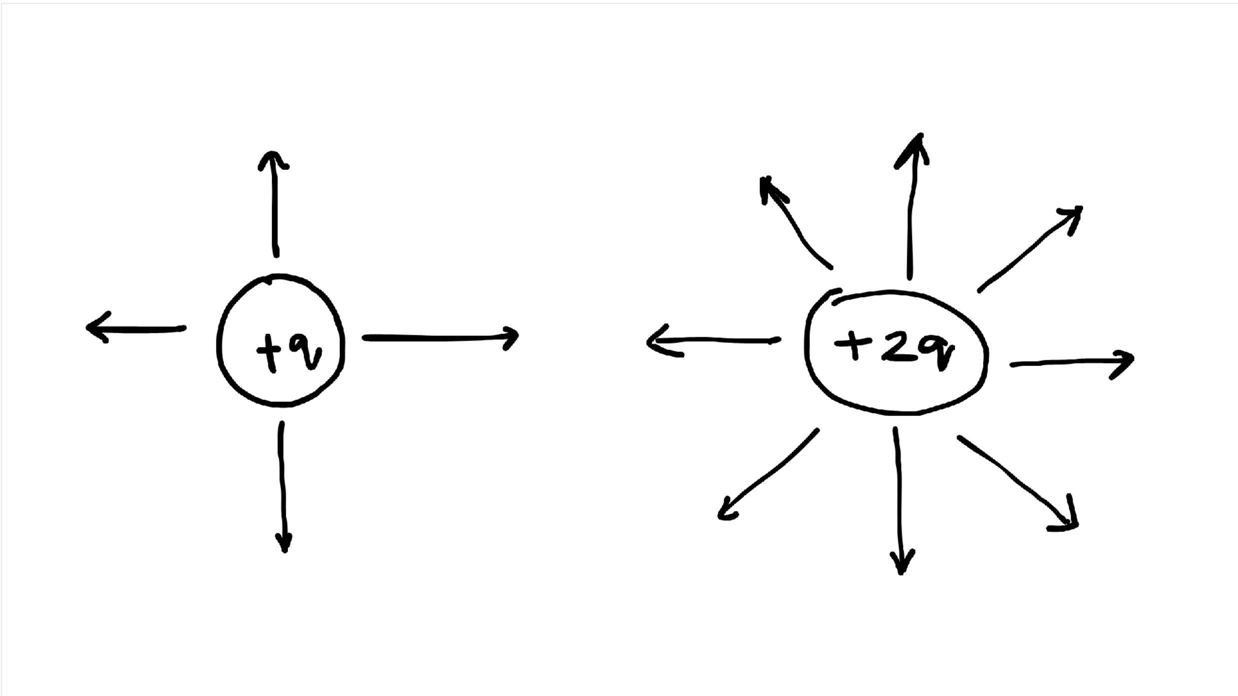
$$\int d\vec{E} = \int \frac{1}{4\pi_0} \frac{\sigma dA}{r^2} \hat{r}$$

4. **Volume Charge**:

$$\int d\vec{E} = \int \frac{1}{4\pi_0} \frac{\rho dV}{r^2} \hat{r}$$

n09-electric_field_lines

- Electric field lines are graphical representation of electric field.
- The **density** of electric field lines gives the **electric field strength** and the **direction** of field lines tells about the direction of Electric field.
- Electric field lines always **originate at** $(+)$ **and terminate at** $(-)$. If 4 lines are generated for a test charge $+q$ then 8 lines will be generated for $+2q$.



- The **flux of electric field** through a surface is proportional to the electric field lines passing through that surface.
- This is denoted by $_E$.

- The electric flux through any surface is given by

$$_E = \vec{E} \cdot d\vec{a} \dots (1)$$

n10-gauss_law

Consider a charge at the origin. Calculate the electric flux through a sphere of Radius R centered at the origin.

$$_E = \vec{E} \cdot d\vec{a}$$

$$= \frac{q}{4\pi_0 r^2} \hat{r} \cdot d\vec{a}$$

$$d\vec{a} = r^2 \sin\theta d\theta d\phi \hat{r}$$

$$_E = \vec{E} \cdot d\vec{a}$$

$$= \frac{q}{4\pi_0 r^2} \hat{r} \cdot r^2 \sin\theta d\theta d\phi \hat{r}$$

$$= \frac{q}{4\pi_0} \int \sin\theta d\theta d\phi (\hat{r} \cdot \hat{r})$$

$$_E = \frac{q}{4\pi_0} \int_0^{\pi} \sin\theta d\theta \int_0^{2\pi} d\phi$$

$$= \frac{q}{4\pi_0} - \cos\theta_0^{\pi} \phi_0^{2\pi}$$

$$= \frac{q}{4\pi_0} (1 - (-1)) 2\pi$$

$$= \frac{q}{4\pi_0} 22\pi$$

$$_E = \frac{q}{_0}$$

$$_E = \vec{E} \cdot d\vec{a} = \frac{q_{\text{en}}}{_0}$$

From the above result, we can conclude that the electric flux through any surface depends on the magnitude of the charge enclosed by the surface. Therefore, the electric flux through any arbitrary surface is equal to $\frac{1}{_0}$ times Charge enclosed by the surface. (This is **Gauss's Law**).

Suppose if there are number of charges scattered at the centre then the electric field is given by the superposition principle.
Now the electric flux through that surface is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \cdots + \vec{E}_n$$

$$_E = \frac{_i\, q_i}{_0}$$

$$_E = \vec{E} \cdot d\vec{a}$$

$$= \left(\;_i \vec{E}_i \right) \cdot d\vec{a}$$

$$_{\text{en}} = \;_i\, q_i$$

Here $_{\text{en}} = \;_i\, q_i$ is **total charge enclosed**.

Limitations: Although Gauss law holds good to find the electric field, it is a useful tool only if the charge distribution is symmetric.

NOTE: If charge is **spherically symmetric** then electric field is calculated taking **sphere** as gaussian surface.

**Question: Find the electric field outside and inside a uniformly charged solid sphere of radius R and total charge .

Electric Field Outside ($r > R$)

$$E_{\text{outside}} = \frac{k}{r^2}$$

Construct a Gaussian Surface (a sphere of radius r) **at a distance** r **from the centre of sphere**.

$$\vec{E} \cdot d\vec{a} = \frac{q_{\text{en}}}{_0}$$

Since $\vec{E}$ and $d\vec{a}$ are parallel and $\vec{E}$ is constant on the Gaussian surface:

$$|\vec{E}|\ |d\vec{a}| = \frac{}{_0}$$

$$|\vec{E}| \int da = \frac{}{_0}$$

$$|\vec{E}|(4\pi r^2) = \frac{}{_0}$$

$$|\vec{E}| = \frac{}{4\pi _0 r^2}$$

$$\vec{E} = \frac{}{4\pi _0 r^2}\hat{r}$$

The field is same as that of if the charge concentrated at the centre.
Electric Field Inside ($r < R$)

$$\vec{E} \cdot d\vec{a} = \frac{q_{\text{en}}}{_0}$$

The charge enclosed q_{en} is:

$$q_{\text{en}} = \rho \cdot \frac{4}{3}\pi r^3$$

where $\rho = \frac{}{\frac{4}{3}\pi R^3}$ is the volume charge density.

$$|\vec{E}|4\pi r^2 = \frac{q_{\text{en}}}{0} = \frac{\rho\frac{4}{3}\pi r^3}{0}$$

$$|\vec{E}| = \frac{\rho r}{3_0}$$

Substituting ρ:

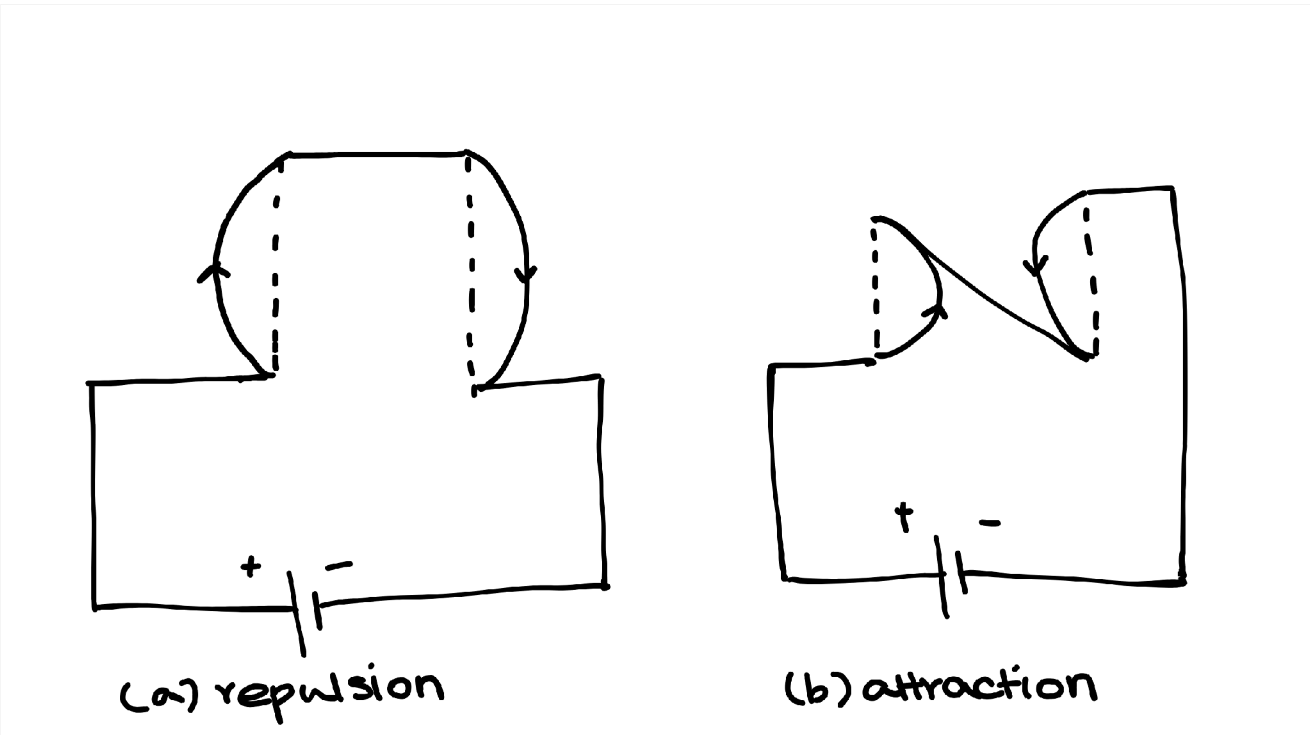
$$\vec{E} = \frac{r}{3_0 \quad \frac{4}{3}\pi R^3}\hat{r} = \frac{r}{4\pi_0 R^3}\hat{r}$$

Graph of Electric Field (E) vs Distance (r)
The graph shows $\vec{E} \quad r$ inside ($r \leq R$) and $\vec{E} \quad 1/r^2$ outside ($r > R$).

Divergence and Curl of Electric Field

- $$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{0}$$
- $$\vec{\nabla} \times \vec{E} = 0$$
- n02-divergence
- n03-curl

n11-magnetic_force



- If current is in same direction, attraction takes place

- If current is in antiparallel directions, repulsion takes place
- The force that accounts for the repulsion in antiparallel currents and attraction due to parallel currents is known as magnetic force.

n12-lorentz_force

- When a test charge q moving with velocity $\vec{v}$ in the presence of magnetic field $\vec{B}$, then the force on q is given by

$$\vec{F}_{\text{ma}} = q(\vec{v} \times \vec{B})$$

  This is called Lorentz force.
- If electric and magnetic fields both are present then the electromagnetic force on q is

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

- The important difference between the electric and magnetic forces is that electric forces do work on charged particles by while the magnetic forces do not do work on moving particle.
- The distance travelled by charged particle in a time dt is given

$$dl = v\,dt$$

- Now the work done is

$$W_{\text{ma}} = \int \vec{F}_{\text{ma}} \cdot dl$$

$$W_{\text{ma}} = \int q(\vec{v} \times \vec{B}) \cdot v\,dt$$

$$W_{\text{ma}} = 0$$

This is the direct consequence of lorentz force law.

- thus the effect of the magnetic force is to change the direction of the particle.

Continuity equation

$$\nabla \cdot \vec{} + \frac{\partial \rho}{\partial t} = 0$$

- where $\vec{}$ is current charge density
- $\frac{\partial \rho}{\partial t}$ is volume charge density
- For steady currents, $\frac{\partial \rho}{\partial t} = 0$, $\nabla \cdot \vec{} = 0$

divergence and curl of magnetic field

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \rightarrow$$

- divergence of magnetic field is always zero which simplifies the non-existence of monopoles.

n13-ampere's_law

- the curl of magnetic field is proportional to the component of current density $\rightarrow$ along the axis.

$$\nabla \times \vec{B} \quad \rightarrow$$

$$\nabla \times \vec{B} = \mu_0 \rightarrow$$

- This is the differential form of ampere's law
- This can be converted to integral form using Stoke's law, as follows.
- Start with the differential form:

$$\nabla \times \vec{B} = \mu_0 \rightarrow$$

- Apply Stokes' theorem:

$$_{\partial} \vec{B} \cdot d\vec{l} = \int (\nabla \times \vec{B}) \cdot d\vec{A}$$

- Substitute the differential form into this:

$$_{\partial} \vec{B} \cdot d\vec{l} = \mu_0 \int \rightarrow \cdot \, d\vec{A}$$

- Surface integral of current density gives the total current through the surface:

$$_{\text{enclosed}} = \int \rightarrow \cdot \, d\vec{A}$$

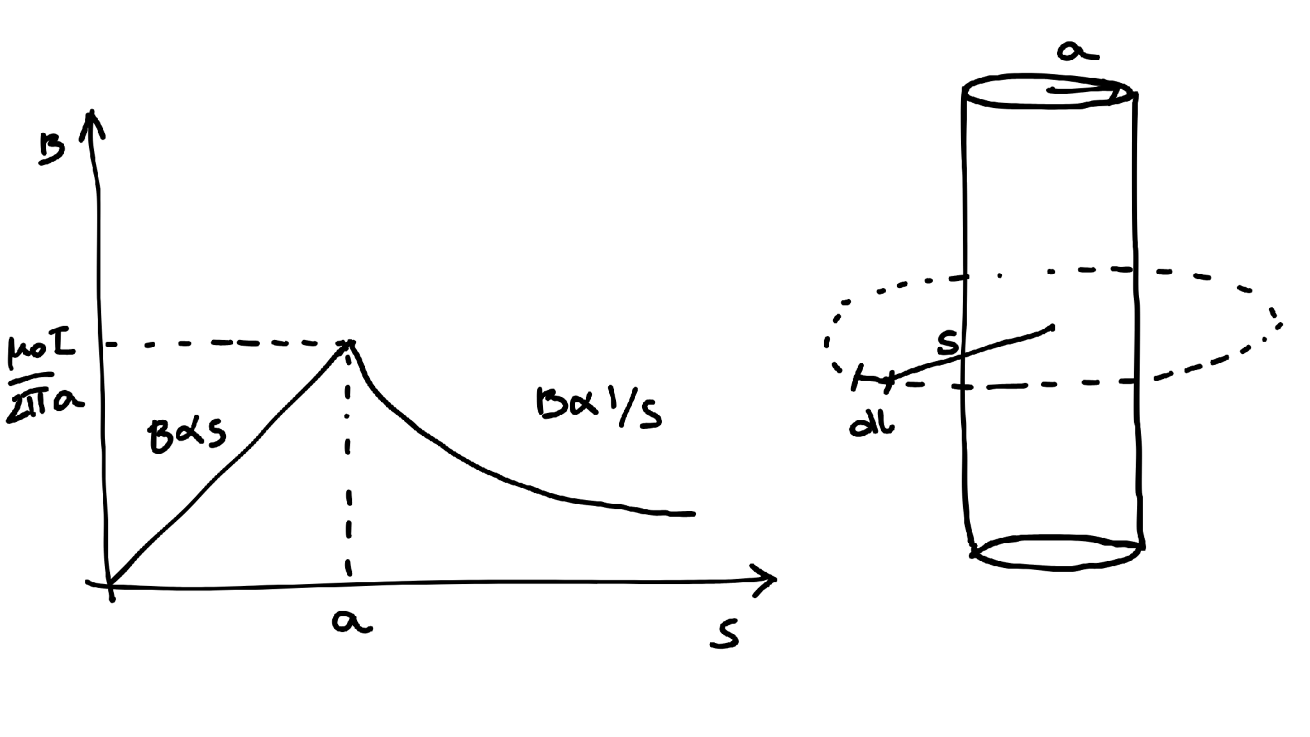
- Hence, the integral form becomes:

$$\boxed{_{\partial} \vec{B} \cdot d\vec{l} = \mu_{0\text{enclosed}}}$$

- This is only applicable for closed loops

Application
Find the magnetic force at a distance 's' from a long straight wire carrying a steady current 'i'. radius of straight wire is 'a'

Let us construct an amperian loop from a distance 's' from an wire consisting of element $d\vec{l}$. From right hand rule we know that magnetic field is circumferential around the wire as shown in figure.



From ampere's law,

$$\vec{B} \cdot d\vec{l} = \mu_{0\text{enc}}$$

$$B\, d\vec{l} = \mu_0 i$$

$$B \cdot (2\pi s) = \mu_0 i$$

$$B = \frac{\mu_0 i}{2\pi s}$$

for a point inside the wire,

$$\vec{B} \cdot d\vec{l} = \mu_{0\text{enc}}$$

$$B \cdot (2\pi s) = \mu_0 (\pi s^2)$$

$$B \cdot (2\pi s) = \mu_0 (\pi s^2)$$

$$B \cdot (2\pi s) = \mu_0 \left(\frac{}{\pi a^2} \right) (\pi s^2)$$

$$B \cdot (2\pi s) = \mu_0 \frac{s^2}{a^2}$$

$$B_{inside} = \frac{\mu_0 s}{2\pi a^2}$$

On the wire (s=a)

$$B = \frac{\mu_0}{2\pi a}$$

n14-ohm's_law

Ohms Law

- In order to sustain the current, we have to apply a force on charges.
- for most of the materials, the current density is proportional to force per unit charge.

$$\to \vec{f} \quad ,\text{here } \vec{f} = \frac{\vec{E}}{\nu}$$

$$\to = \sigma \vec{f}$$

- where σ is the proportionality constant called conductivity. It varies from material to material.
- the reciprocal of conductivity is known as resistivity

$$\rho = \frac{1}{\sigma}$$

- In general, the forces on a moving charge are electromagnetic in nature. therefore,

$$\to = \sigma \left\{ q(\vec{E} + \vec{v} \times \vec{B}) \right\}$$

since $\vec{F}_{em} = q\vec{E} + q(\vec{v} \times \vec{B})$

$$\to = \sigma(\vec{E} + \vec{v} \times \vec{B})$$

If the velocity of charge is small, then the second term can be neglected

$$\to = \sigma \vec{E}$$

- This is called ohm's law.

Question

A cylindrical resistor of cross sectional area A and length L is made from a material with conductivity σ. If the potential is constant over each end and the potential difference between the ends is V, what current flows?

Answer

$$\sigma = \frac{}{A}$$
$$= A$$

but,

$$\to = \sigma \vec{E}$$
$$= \sigma E A$$

also,

$$\vec{E} = \frac{V}{L}$$

$$= \frac{\sigma V A}{L}$$

n15-electromotive_force

Electromotive Force

- EMF is defined as the voltage produced by the source of electric field such as battery or time varying magnetic field.
- there are 2 driving forces in a circuit.

$$\vec{f} = \vec{f}_s + \vec{E} \ldots (i)$$

where

- $\vec{f}_s \to$ force due to source connected to a circuit.
- $\vec{E} \to$ electrostatic field
- an EMF in a circuit is given by the line integral of this resultant force. i.e.

$$\mathcal{E} = \vec{f} \cdot d\vec{l}$$

$$\mathcal{E} = (\vec{f}_s + \vec{E}) \cdot d\vec{l} \quad \ldots (\text{from i})$$

$$\mathcal{E} = \vec{f}_s \cdot d\vec{l} + \vec{E} \cdot d\vec{l}$$

$$\mathcal{E} = \vec{f}_s \cdot d\vec{l} \quad \ldots (\text{ii})$$

- for any ideal source, $\vec{f}_s = 0$ ($\text{since } \sigma = \infty, \quad \vec{\ } = \sigma \vec{f}$) therefore from equation (i)

$$\mathcal{E} = -\vec{f}_s \quad \ldots (\text{iii})$$

- Now,

$$V = -\int_a \vec{E} \cdot d\vec{l}$$

$$V = \int_a \vec{f}_s \cdot d\vec{l} \quad \ldots (\text{from iii})$$

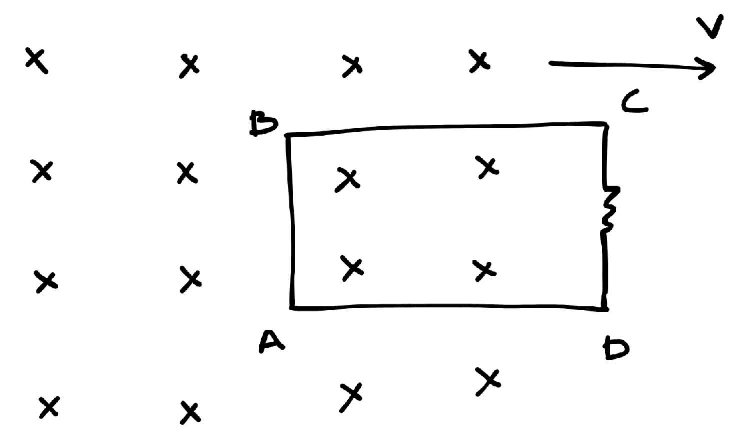
- If we extend the integral over an entire loop.

$$= \vec{f}_s \cdot d\vec{l}$$
$$= \mathcal{E}$$

- Therefore the function of battery is to establish and maintain the voltage in the loop

n16-motional_emf

- An EMF induced by motion in the presence of magnetic field is known as **motional EMF**
- It arises when we move a wire or loop in a magnetic field.
- In general, generators exploit the motional EMF.
- A primitive model of generator is shown in the figure in which the rectangular loop ABCD moves towards right with velocity $\vec{V}$



- When the loop is moving away from the magnetic field region, electrons on the left side of the wire experience a magnetic force which makes the current flow in clockwise direction in the loop.
- Therefore,

$$\mathcal{E} = \vec{f} \cdot d\vec{l}$$

$$= q\vec{v} \times \vec{B} \cdot d\vec{l}$$

$$\mathcal{E} = vBh \quad \ldots(i)$$

- Here, the contribution is only from the left side AB of the loop and the contribution from horizontal segments BC and AD is 0
- A motional EMF can be expressed in the form of magnetic flux of magnetic field in the loop.
- The magnetic flux is defined as

$$= \int \vec{B} \cdot \vec{d\mathrm{a}}$$

$$= B \int \mathrm{da}$$

$$= BA$$

$$= Bhx$$

- Now, the rate of change of flux

$$\frac{d}{dt} = Bh\frac{dx}{dt}$$

$$\frac{d}{dt} = Bhv$$

- Since the loop is moving away from the magnetic field, the change in magnetic flux is

$$\frac{d}{dt} = -Bhv \quad \dots (ii)$$

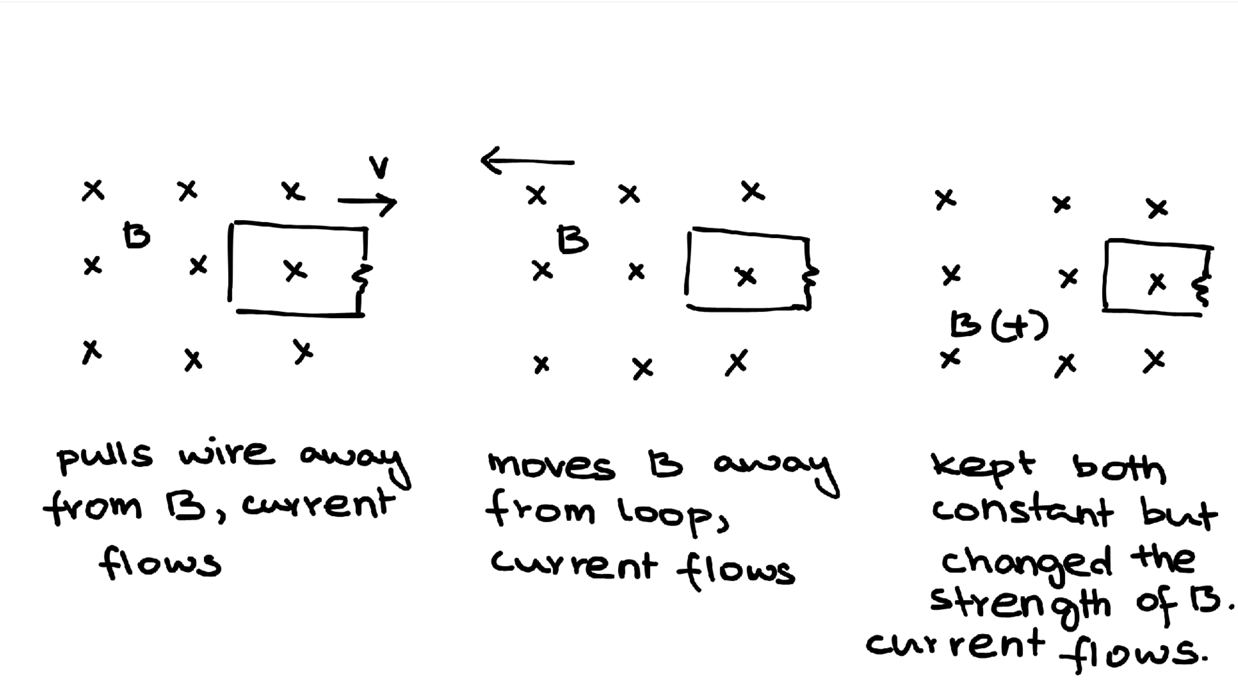
- So from (i) and (ii)

$$\mathcal{E} = -\frac{d}{dt}$$

- That is, Induced EMF is negative the rate of change of magnetic flux.
- This is known as flux rule for motional EMF.

n17-electromagnetic_induction

- The phenomenon of generation of induced EMF in a closed circuit due to change in magnetic field is known as electromagnetic induction.
- The operation of motors, generators and transformers uses the principle of emi.
- In the year 1831, Michael Faraday performed a series of experiments as follows.



- In the first experiment, motional EMF is induced which is given by

$$\mathcal{E} = -\frac{d}{dt}$$

- In the second and third experiments, charges are at rest and the reason for flow of current is induced electric field. This electric field induces EMF in the loop. Faraday empirically found that induced EMF is equal to rate of change of magnetic flux linked in the circuit.

$$\mathcal{E} = \vec{E} \cdot d\vec{l} = -\frac{d}{dt}$$

$$\vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

- This is the integral form of Faraday's law
- Now, using stoke's theorem.

$$\int (\nabla \times \vec{E}) \cdot d\vec{a} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

- assuming surface is stationary

$$\int (\nabla \times \vec{E}) \cdot d\vec{a} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{a}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

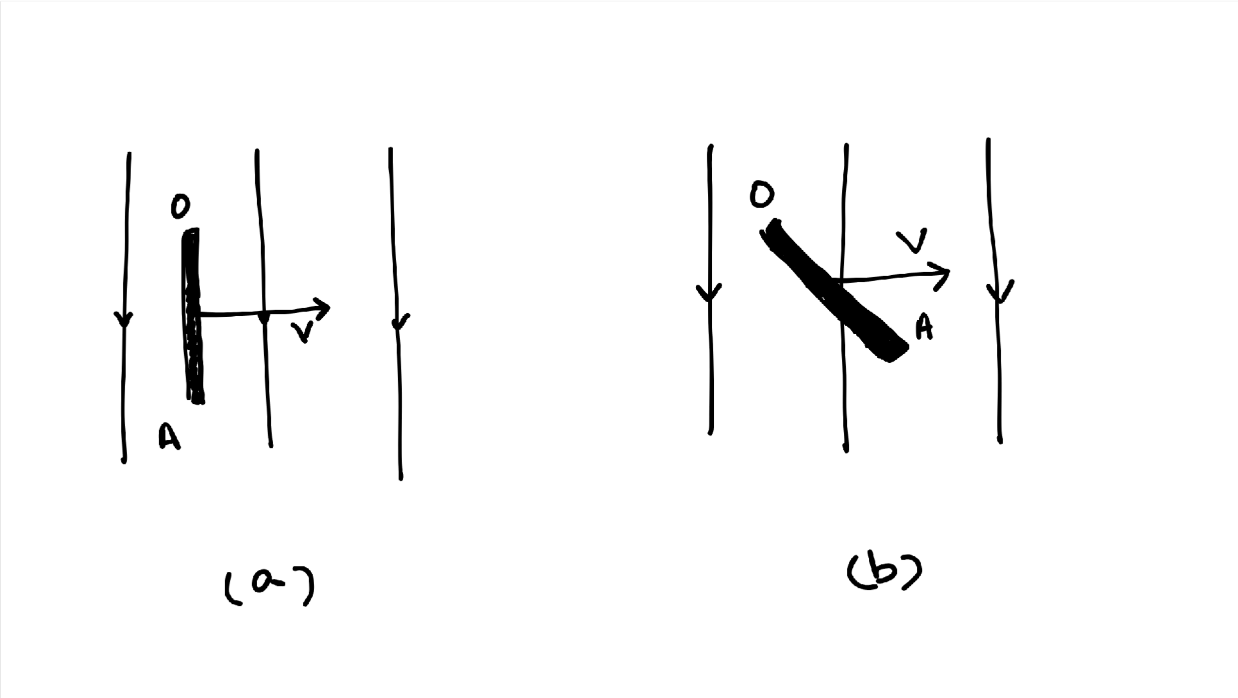
- This is the differential form of Faraday's law
- Faraday Summed up the result of all the three experiments and gave them as 2 laws.
    - Faraday's first law: It postulates that whenever the magnetic flux linked with a closed circuit changes, an emf is always induced in it.
    - Faraday's second law: It postulates that the induced emf is directly proportional to the flux or flux linkage (flux number of turns).

n18-dynamically_induced_emf

Dynamically Induced EMF

- Dynamically Induced EMF is produced by the movement of conductor in a uniform magnetic field.

- The figure shows uniform magnetic field of flux density $'B'$ tesla



- Let $'OA'$ be a conductor moving in a direction shown in & and cuts the flux at right angles.
- If
  - $'l'$ = length of conductor
  - $'v'$ = velocity of conductor
  - $'dx'$ = distance travelled by conductor in time $'dt'$
- Then area swept by conductor is given by
  Area = $'ldx'$
- Now,

$$\phi = \int B \cdot ds = BA$$

$$\phi = Bldx$$

- Therefore

$$\frac{d\phi}{dt} = Bld\frac{x}{dt}$$

$$\frac{d\phi}{dt} = BLV$$

- From the definition of EMF

$$= \frac{d\phi}{dt}$$

$$= BLV$$

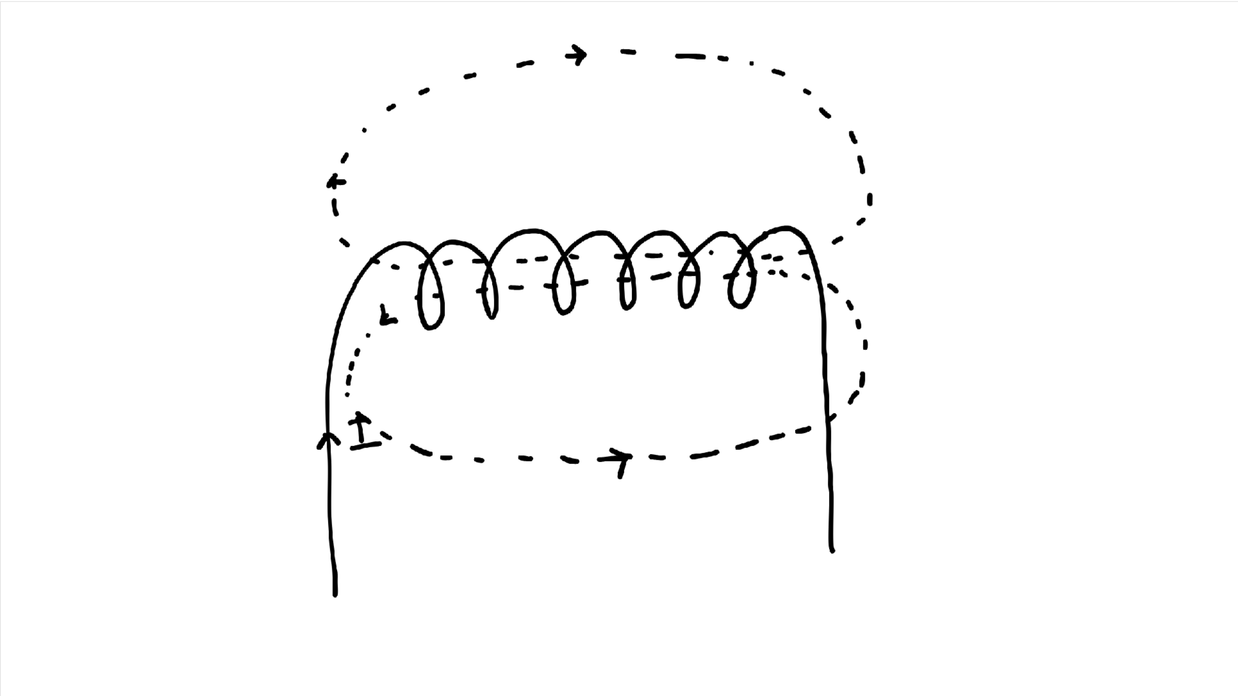
- If the conductor moves making an angle $'\theta'$ with magnetic flux as shown in figure (b), then induced emf

$$= BLV \sin\theta$$

n19-self-induced_emf

Self-induced EMF

- The property of the coil which enables it to induce an EMF whenever there is change in a current in a circuit is called self-induction.
- The EMF Induced is called self Induced EMF.
- consider a coil of $'N'$ turns



- Let '' be the current passing through the coil and $'\phi'$ be the flux linked with the coil as shown in the figure.
- If the current passing through the coil changes then the magnetic flux linked withe the coil also changes
i.e.

$$\frac{\phi}{} = cnst \quad (i)$$

- Here flux linkage $\phi_s = N\phi$
- From faraday's second law, we have

$$= -\frac{d}{dt}(N\phi)$$

$$= -N\frac{d\phi}{dt} \quad (ii)$$

- Now,

$$\frac{d\phi}{dt} = \frac{\phi}{} \times \frac{d}{dt} \quad (iii)$$

from (ii) and (iii)

$$= -N \frac{\phi}{} \cdot \frac{d}{dt}$$

$$= -\left(\frac{N\phi}{}\right) \frac{d}{dt}$$

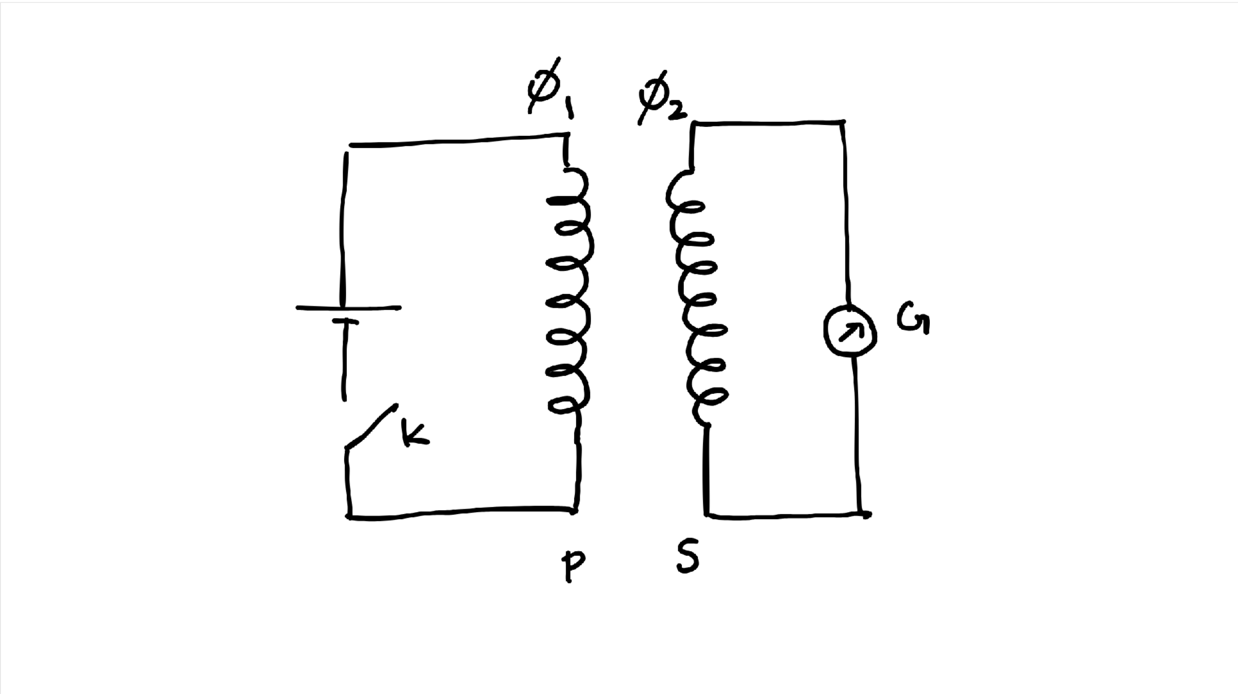
$$= -L \frac{d}{dt}$$

- The negative sign in the above expression opposes the change i.e. when current is increasing it opposes the increase of current and when current is decreasing in the circuit the current prevents the decay.

n20-mutually_induced_emf

Mutually Induced EMF

- the phenomenon of generation of induced EMF in a circuit due to the flow of current in a neighboring circuit is called mutual induction.
- the EMF generated is known as mutually induced EMF.
- Consider 2 coils $'P'$ and $''$ in such a way that the coil $'P'$ is connected to a cell through key $'k'$ and the coil $''$ is connected to a galvanometer as shown in figure.



- Let
  - $\phi_1 =$ flu in the coil due to the current in
  - $\phi_2 =$ flu in the coil due to ϕ_1 in
- When the key $'k'$ is closed, there is a sudden kick in galvanometer and when the key is opened again there is ic deflection in the galvanometer in opposite direction

- This shows that there is a current flow in coil '' even though it is not connected physically to the coil 'P'
- The coils which exhibit this property are said to have mutual inductance.
- Let
  - $\phi_1 =$ flu in the coil due to the current in
  - $\phi_2 =$ flu in the coil due to ϕ_1 in
- Now,

$$\phi_1 \quad \phi_2$$

$$\frac{\phi_2}{\phi_1} = k$$

$$\phi_2 = k\phi_1 \quad \ldots(i)$$

- Also

$$\phi_2 = \frac{\phi_2}{1} \times 1$$

$$\phi_2 = k\phi_{11} \quad \text{from } (i)$$

- From Faraday's second law,

$$_s = \frac{d}{dt}(N_2\phi_2)$$

$$_s = -N_2\frac{d\phi_2}{dt}$$

$$_s = -N_2\frac{k\phi_1}{1} \cdot \frac{d}{dt} \quad (\text{from } (i))$$

$$_s = -N_2\frac{\phi_2}{1} \cdot \frac{d_1}{dt} \quad (\text{from } (i))$$

$$_s = -\frac{d}{dt}$$

where

$$= N_2\frac{\phi_2}{1} = \text{ utual nductance}$$

Example
1. A uniform magnetic field '$B(t)$' pointing Straight up fills the circular region as shown in figured. If 'B' is charging with time, what is induced electric field?

n21-electrodynamics

Electrodynamics involves three types of fields:

- **Electric field produced by electric charges**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{_0}$$

$$\nabla \times \mathbf{E} = 0$$

- B field produced by electric current

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} = \mu_0$$

- G field produced by change in magnetic field

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot E = 0$$

n22-maxwell's_equations

The basic laws of electromagnetism including **Gauss law for electricity**, **Gauss law for magnetism**, **Faraday's law of Electromagnetic Induction** and **Ampere's Law in conductors** are collectively known as **Maxwell's equations**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{_0} \quad \ldots(i)$$

$$\nabla \cdot \mathbf{B} = 0 \quad \ldots(ii)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \ldots(iii)$$

$$\nabla \times \mathbf{B} = \mu_0 \quad \ldots(iv)$$

Taking divergence of equation (iii):

$$\nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \left(-\frac{\partial \mathbf{B}}{\partial t}\right)$$

$$0 = -\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B})$$

In the above equation **LHS is zero** since it is representing the curl of $\mathbf{E}$ and **RHS is zero from (ii)**.

Taking **divergence of equation (iv)**:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0(\nabla \cdot)$$

In the above equation **LHS is zero** but **RHS is zero only for steady current**. In other cases $\nabla \cdot \; = -\frac{\partial \rho}{\partial t}$, so a **contradiction arises**. The failure of Ampere's law can also be observed where the **capacitor is charged**.

Consider an **Amperian loop** as shown in the figure for the **flat surface**, which is in the plane of the circuit $_{\text{enc}} =$, but for **balloon shaped surface**, $_{\text{enc}} = 0$ (no

conductive path between the plates).
Thus Ampere's law holds good only when the current is continuous.

Maxwell's correction to Ampere's Law

The contradiction arises as $\nabla \cdot$ is not always zero; however, combining the **continuity equation** and **Gauss Law** we can write as follows:

$$\nabla \cdot \;= -\frac{\partial \rho}{\partial t}$$

$$\nabla \cdot \;= -\frac{\partial}{\partial t} {}_0(\nabla \cdot \mathbf{E})$$

Maxwell replaced ρ by $-{}_0\nabla \cdot \mathbf{E}$ in the continuity equation:

$$\nabla \cdot \;= -{}_0\nabla \cdot \left(\frac{\partial \mathbf{E}}{\partial t}\right)$$

$$\nabla \cdot \;+ {}_0\frac{\partial \mathbf{E}}{\partial t} = 0$$

Maxwell replaced with $+{}_0\frac{\partial \mathbf{E}}{\partial t}$ in Ampere's Law:

$$\nabla \times \mathbf{B} = \mu_0 + \mu_{00}\frac{\partial \mathbf{E}}{\partial t} \quad \ldots(v)$$

The extra term gave additional symmetry to the maxwell's equations and Maxwell named it as **displacement current**.

The **displacement current density** is:

$$_{\mathbf{d}} = {}_0\frac{\partial \mathbf{E}}{\partial t}$$

where $_{\mathbf{d}} \rightarrow$ displacement current density
It is **not a genuine current** in the sense of movement of charges but it has the **units of real current density**.

Taking **area integral on both sides for eqn (v)**:

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \mu_0 \int \cdot d\mathbf{a} + \mu_{00} \int \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a}$$

Using Stokes' theorem and the definition of current:

$$\mathbf{B} \cdot d = \mu_{0\text{enc}} + \mu_{00}\frac{\partial}{\partial t} \int \mathbf{E} \cdot d\mathbf{a} \quad \ldots(vi)$$

For **flat surface**: $_{\text{enc}} =$ and $\frac{\partial \mathbf{E}}{\partial t} = 0$ (Outside capacitor)

$$\mathbf{B} \cdot d = \mu_0$$

For **balloon surface**: $_{\text{enc}} = 0$ (Between capacitor plates)

The electric field $\mathbf{E}$ between the capacitor plates is:

$$E = \frac{\sigma}{_0} = \frac{}{A_0}$$

where is the charge, A is the area of the plates.

The time derivative of $\mathbf{E}$ is:

$$\frac{\partial E}{\partial t} = \frac{1}{A_0}\frac{\partial}{\partial t}$$

Since $\frac{\partial}{\partial t} =$ (the charging current):

$$\frac{\partial E}{\partial t} = \frac{}{A_0}$$

Equation (vi) becomes (for balloon surface, $\mathbf{E}$ is constant over the area A):

$$\mathbf{B} \cdot d = 0 + \mu_{00} \int \frac{}{A_0} d\mathbf{a}$$

$$\mathbf{B} \cdot d = \mu_{00} \frac{}{A_0} \int d\mathbf{a}$$

$$\mathbf{B} \cdot d = \mu_0$$

Both the cases are true and are proved, but in first case it is due to **genuine current** and in second case it is due to **displacement current**.

Maxwell's equations

The complete set of Maxwell's equations in differential form are given by:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 + \mu_{00} \frac{\partial \mathbf{E}}{\partial t}$$

In integral form:

$$\mathbf{E} \cdot d\mathbf{a} = \frac{_{\text{enc}}}{_0}$$

$$\mathbf{B} \cdot d\mathbf{a} = 0$$

$$\mathbf{E} \cdot d = -\frac{\partial}{\partial t} \int \mathbf{B} \cdot d\mathbf{a}$$

$$\mathbf{B} \cdot d = \mu_{0\text{enc}} + \mu_{00} \int \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a}$$

n23-propagation_of_electromagnetic_waves

Maxwell equations in **free space** are given by putting $\rho = 0$ and $= 0$.

$$\nabla \cdot \mathbf{E} = 0 \quad \ldots (i)$$

$$\nabla \cdot \mathbf{B} = 0 \quad \ldots (ii)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \ldots (iii)$$

$$\nabla \times \mathbf{B} = \mu_{00} \frac{\partial \mathbf{E}}{\partial t} \quad \ldots (iv)$$

Taking **curl of eqn (iii)**:
Using the vector identity $\mathbf{A} \times (\mathbf{B} \times) = (\mathbf{A} \cdot)\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})$:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right)$$

$$\nabla(\nabla \cdot \mathbf{E}) - (\nabla \cdot \nabla)\mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B})$$

Using (i) and (iv):

$$\nabla(0) - \nabla^2 \mathbf{E} = -\frac{\partial}{\partial t} \left(\mu_{00} \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$\nabla^2 \mathbf{E} = \mu_{00} \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \ldots (v)$$

Similarly taking **curl of eqn (iv)**:

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla \times \left(\mu_{00} \frac{\partial \mathbf{E}}{\partial t} \right)$$

$$\nabla(\nabla \cdot \mathbf{B}) - (\nabla \cdot \nabla)\mathbf{B} = \mu_{00} \frac{\partial}{\partial t}(\nabla \times \mathbf{E})$$

Using (ii) and (iii):

$$\nabla(0) - \nabla^2 \mathbf{B} = \mu_{00} \frac{\partial}{\partial t} \left(-\frac{\partial \mathbf{B}}{\partial t} \right)$$

$$\nabla^2 \mathbf{B} = \mu_{00} \frac{\partial^2 \mathbf{B}}{\partial t^2} \quad \ldots (vi)$$

for a **general wave**:

$$\nabla^2 f = \frac{1}{V^2} \frac{\partial^2 f}{\partial t^2} \quad \ldots (vii)$$

Comparing equations (v), (vi), and (vii):

$$\frac{1}{V^2} = \mu_{00}$$

$$V^2 = \frac{1}{\mu_{00}}$$

$$V = \frac{1}{\sqrt{\mu_0{}_0}}$$

Substituting the values of $\mu_0 = 4\pi \times 10^{-7}\ /\mathrm{m}$ and ${}_0 = 8.85 \times 10^{-12}\ /\mathrm{m}$:

$$V = 2.8 \times 10^{8}\ \mathrm{m/s}$$

Hence the EMW propagates with the **velocity of light** in free space.